# Implementation of Timer using 8051

**LAB # 06**

**Spring 2024**

**CSE-307L Microprocessor Based System Design Lab**


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"On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Submitted to:

**Dr. Asif Ali Khan**

Date:

**29th March 2024**


## Department of Computer Systems Engineering

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## Task 1:

Use timer to Generate a delay of 15ms

## Calculation:

For 15ms delay, subtract $15{,}000_{dec}$ from $FFFF_{hex}$

Hence, we get the starting value of the 16-bit counter, i-e

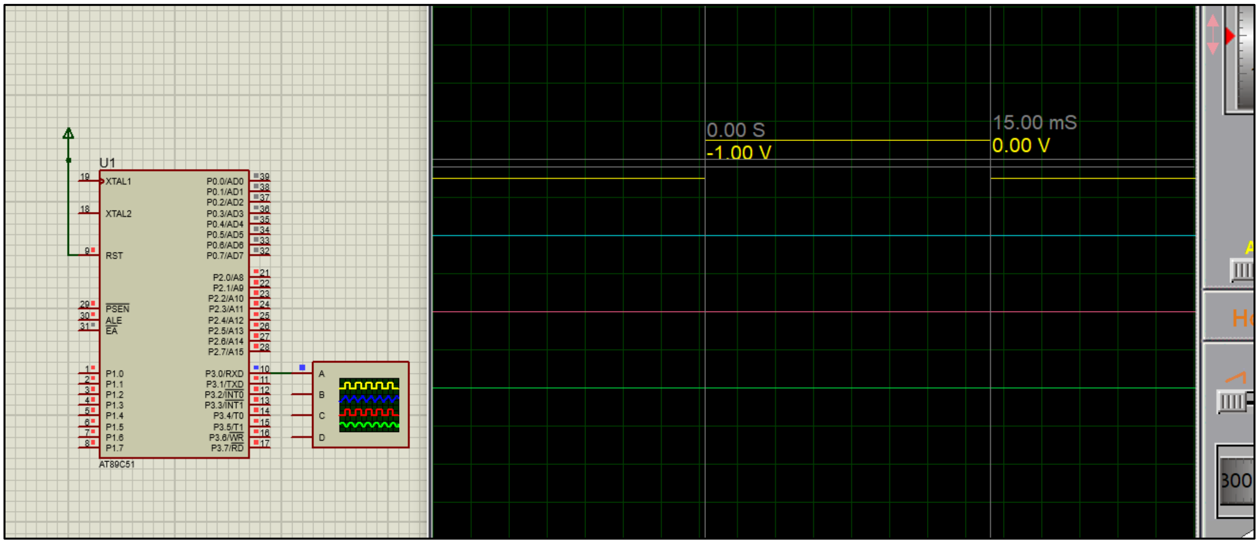
$FFFF_{hex} - 15{,}000_{dec} = \mathbf{C567_{hex}}$

## Code:

task1.c

```c
#include<reg51.h>
#include<stdio.h>
sbit pin = P3^0;

void start_timer0(){
  TR0 = 1;
}

void timer0() interrupt 1{
  TH0 = 0xC5; //15ms delay
  TL0 = 0x67;
}

void start_timer(){
  TMOD = 0x01;
  IE = 0x82;
}

void main(void){

      TR0 = 1;
      start_timer();

      while(1){
        while(TF0 == 0);
        pin = ~pin;

      }
}
```

**Proteus Simulation:**




## Task 2:

Use timer to Generate a delay of 50ms

## Calculation:

For 50ms delay, subtract $50{,}000_{dec}$ from $FFFF_{hex}$

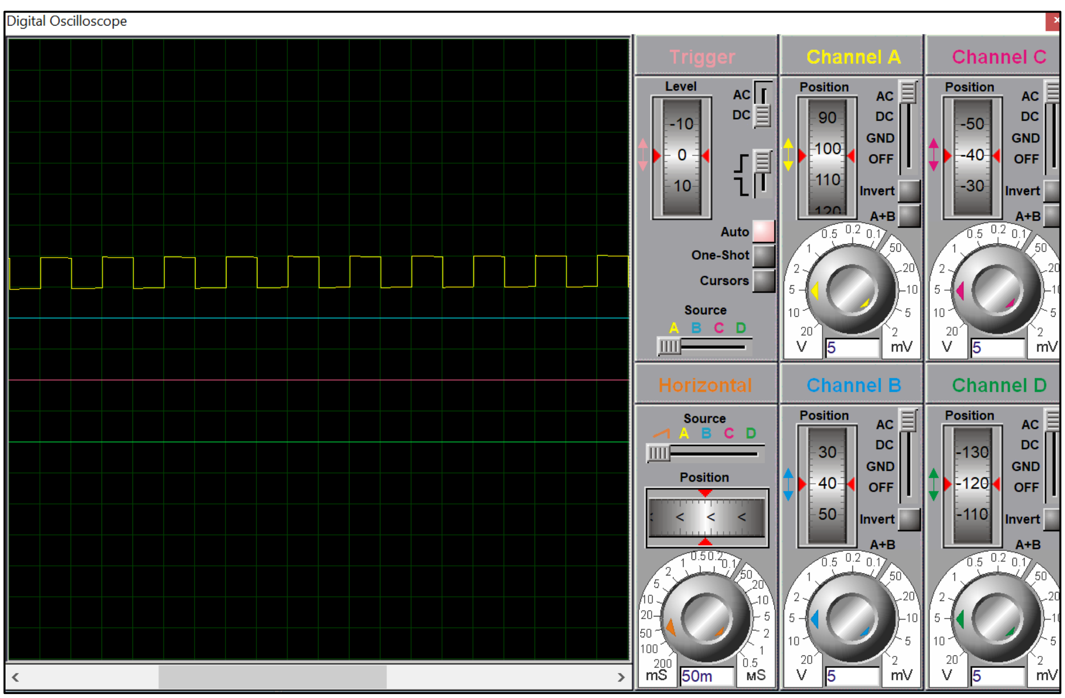
Hence, we get the starting value of the 16-bit counter, i-e

$$FFFF_{hex} - 50{,}000_{dec} = \mathbf{3CAF_{hex}}$$

**Code:**

```
task2.c
#include<reg51.h>
#include<stdio.h>
sbit pin = P3^0;

void start_timer0(){
  TR0 = 1;
}

void timer0() interrupt 1{
  TH0 = 0x3C; //50ms delay
  TL0 = 0xAF;
}

void start_timer(){
  TMOD = 0x01;
  IE = 0x82;
}

void main(void){

      TR0 = 1;
      start_timer();

      while(1){

        while(TF0 == 0);
        pin = ~pin;

      }
```

**Proteus Simulation:**



## Task 3:

Use timer to Generate a delay of 200ms

### Calculation:

We cannot directly create 200ms delay as it is out of the bounds of a 16-bit timer(Max Delay = 65ms approximately). Hence, we will create 50ms delay first and then wait for it 4 times i-e

50ms x 4 = **200ms**

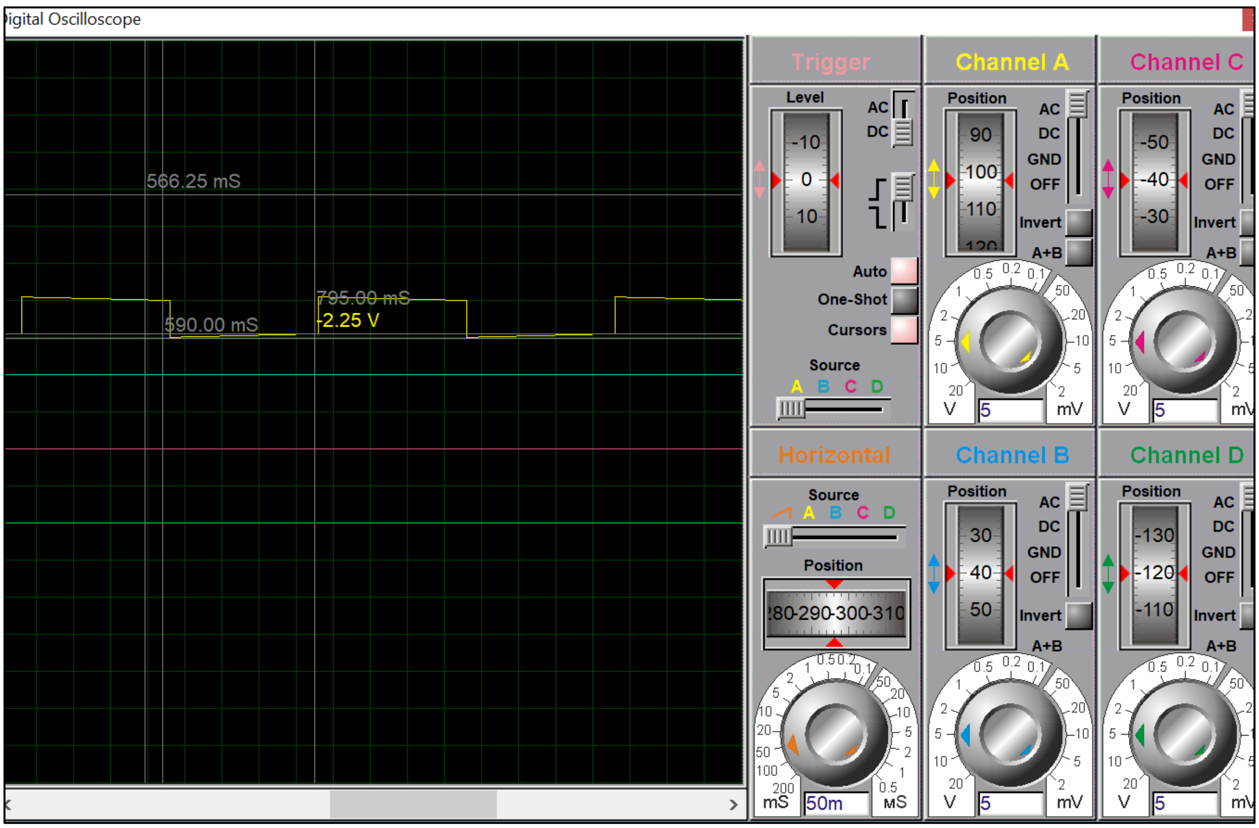
### Code:

task3.c

```c
#include<reg51.h>
#include<stdio.h>

sbit pin = P3^0;
int count;

void timer0() interrupt 1{

  if(count == 4){
    pin = ~pin;
    count = 0;
  }

    TH0 = 0x3C;
    TL0 = 0xAF;
}

void start_timer(){
  TMOD = 0x01;
  IE = 0x82;
}

void main(void){
      TR0 = 1;
      start_timer();
```

```c
void main(void){
        TR0 = 1;
        start_timer();

        while(1){

          while(TF0 == 0);
          count++;
        }
}
```

**Proteus Simulation:**



## Task 4:

Use timer to Generate a delay of 200µs

## Calculation:

For 200µs delay, subtract $200_{dec}$ from $FFFF_{hex}$

Hence, we get the starting value of the 16-bit counter, i-e

$$FFFF_{hex} - 200_{dec} = \mathbf{FF37_{hex}}$$

## Code:

task4.c

```c
#include<reg51.h>
#include<stdio.h>
sbit pin = P3^0;

void timer0() interrupt 1{
     TH0 = 0xFF;
     TL0 = 0x37;
}

void start_timer(){
   TMOD = 0x01;
   IE = 0x82;
}

void main(void){
        TR0 = 1;
        start_timer();

        while(1){
          while(TF0 == 0);
              pin = ~pin;
        }
}
```

**Proteus Simulation:**